

IE-28: Recovered Work on Positive Diagonal Preconditioning

Constructive low-stage proofs, an all-stage local theorem,
and reproducible algebraic and numerical checks

12 September 2026

Completion status. This archive is **not a complete solution of IE-28**. It proves existence for every two-node and every three-node set, and for every stage count when the nodes are sufficiently close to a common positive point. It also supplies an exact certificate for one four-node set. It does not prove existence, or exhibit nonexistence, for all admissible node sets at arbitrary stage count.

Abstract

The interrupted calculation left recoverable research notes and code fragments, but no surviving files or completed all-stage proof. This document reconstructs and checks the defensible mathematics. A Vandermonde/Cauchy reformulation gives exact polynomial equations for the diagonal. A logarithmic-derivative parametrization yields an elementary two-stage formula and a constructive intermediate-value proof for arbitrary three-stage nodes. A confluent determinant identity and a nonsingular triangular coefficient map give a local existence theorem in every dimension. An independently replayable rational/algebraic certificate verifies an equally spaced four-stage example. High-precision experiments are reported separately from theorems. No claim of priority or literature novelty is made for these reconstructed results.

1 Target, provenance, and scope

The URL in the request points to **IE-28**, entitled “Positive diagonal preconditioning with a nilpotent stiff limit,” notwithstanding the different visible link label. The target is the following positive-distinct-node formulation in the repository [1]. Given

$$s \geq 2, \quad 0 < c_1 < \cdots < c_s \leq 1, \quad \ell_j(t) = \prod_{k \neq j} \frac{t - c_k}{c_j - c_k},$$

set

$$A_{ij} = \int_0^{c_i} \ell_j(t) dt. \tag{1}$$

The requested assertion is that some real diagonal $D = \text{diag}(d_1, \dots, d_s)$ with every $d_i > 0$ satisfies

$$(I - D^{-1}A)^s = 0. \tag{2}$$

The original conjecture appears in van der Houwen and de Swart, Section 3.2.1, page 46 [2]. The 2025 treatment discusses the fixed-diagonal stiff-limit construction and explicitly notes the lack of a general existence guarantee in Section 2.2.3, page A439 [3]. These references identify the target; the mathematical claims below have their own proofs.

Neither a nilpotent $A - D$ nor a sweep-dependent sequence of diagonals establishes (2). Likewise, allowing a triangular preconditioner changes the problem. No ordering of d_i is required here. A node at zero is excluded: then A has a zero row, so $I - D^{-1}A$ has eigenvalue one for every invertible positive diagonal D .

The supplied output logs were regenerated for this archive. They are not represented as recovered original files. The account in `notes/recovery.md` records this distinction.

2 Exact reformulations in every dimension

2.1 Integration and differentiation on polynomial spaces

Put

$$C = \text{diag}(c_1, \dots, c_s), \quad V = (c_i^{k-1})_{i,k=1}^s, \quad \Lambda = \text{diag}(1, \dots, s).$$

Integrating a polynomial of degree at most $s-1$ gives

$$A = CV\Lambda^{-1}V^{-1}, \quad \det A = \frac{\prod_{i=1}^s c_i}{s!} > 0. \quad (3)$$

In particular, A is invertible. Let

$$\mathcal{P}_s^0 = \{p \in \mathbb{R}[t] : \deg p \leq s, p(0) = 0\}.$$

For $p \in \mathcal{P}_s^0$, write $p(c) = (p(c_1), \dots, p(c_s))^T$. Since p' has degree at most $s-1$, the definition of A implies

$$Ap'(c) = p(c), \quad A^{-1}p(c) = p'(c). \quad (4)$$

These identities will produce an eigenvalue equal to one without solving a characteristic polynomial.

Rescaling all nodes by $\alpha > 0$ multiplies A by α . Thus D is a solution for c exactly when αD is a solution for αc . The upper bound $c_s \leq 1$ is a normalization, not an additional obstruction.

2.2 An explicit Cauchy-type matrix

Define $\delta_i = d_i/c_i$ and $\Delta = \text{diag}(\delta_1, \dots, \delta_s)$. Let

$$w_i = \prod_{j \neq i} (c_i - c_j)^{-1}, \quad W = \text{diag}(w_1, \dots, w_s).$$

The usual polynomial differentiation formula, or direct differentiation of the Lagrange basis, gives

$$B = W(V\Lambda V^{-1})W^{-1}, \quad B_{ij} = \begin{cases} 1 + \sum_{k \neq i} \frac{c_i}{c_i - c_k}, & i = j, \\ \frac{c_i}{c_i - c_j}, & i \neq j. \end{cases} \quad (5)$$

For completeness, $V\Lambda V^{-1} = I + CL$, where L differentiates polynomials of degree at most $s-1$ in their value representation. For $i \neq j$, $L_{ij} = w_j/[w_i(c_i - c_j)]$; its diagonal is $L_{ii} = \sum_{k \neq i} (c_i - c_k)^{-1}$. This proves (5). Because W commutes with Δ ,

$$W(A^{-1}D)W^{-1} = B\Delta. \quad (6)$$

Notice that B need not have positive diagonal entries or positive principal minors. Positivity arguments valid only for such matrices cannot simply be applied to this reformulation.

Proposition 2.1. *For a positive diagonal D , the following conditions are equivalent:*

$$(I - D^{-1}A)^s = 0; \quad (7)$$

$$\det(\lambda I - B\Delta) = (\lambda - 1)^s; \quad (8)$$

$$\sum_{\substack{J \subseteq \{1, \dots, s\} \\ |J|=k}} \det B[J, J] \prod_{j \in J} \delta_j = \binom{s}{k}, \quad k = 1, \dots, s. \quad (9)$$

In particular every solution satisfies

$$\prod_{i=1}^s \delta_i = \frac{1}{s!}. \quad (10)$$

Proof. Let $N = I - D^{-1}A$. Then $A^{-1}D = (I - N)^{-1}$. If N is nilpotent, this inverse is $I + N + \dots + N^{s-1}$, hence has only unit eigenvalues. Conversely, if $A^{-1}D$ has characteristic polynomial $(\lambda - 1)^s$, Cayley–Hamilton makes $A^{-1}D - I$ nilpotent, and its inverse also differs from I by a nilpotent matrix. Equation (6) proves equivalence with (8). Expanding a characteristic polynomial in principal minors proves (9). Finally $\det B = s!$ by (5), giving (10). $\square$

An alternative determinant identity, without computing minors, is

$$\det [c_i^{k-1}(1 - zk\delta_i)]_{i,k=1}^s = \det(V)(1 - z)^s. \quad (11)$$

Indeed, the left side divided by $\det V$ is $\det(I - z\Delta V\Lambda V^{-1})$. These exact coefficient equations are preferable to interpreting a small floating-point spectral radius as proof of nilpotence.

2.3 A useful positive parametrization

Lemma 2.2. *Suppose $p \in \mathcal{P}_s^0$ and $p(c_i)/p'(c_i) > 0$ for every i . Then*

$$d_i = \frac{p(c_i)}{p'(c_i)}$$

defines a positive diagonal for which DA^{-1} , and hence $A^{-1}D$, has eigenvalue one. In particular, for

$$p(t) = t \prod_{j=1}^{s-1} (t + r_j), \quad r_j > 0, \quad (12)$$

we have

$$d_i = \left(\frac{1}{c_i} + \sum_{j=1}^{s-1} \frac{1}{c_i + r_j} \right)^{-1}, \quad \delta_i = \left(1 + \sum_{j=1}^{s-1} \frac{c_i}{c_i + r_j} \right)^{-1}. \quad (13)$$

These satisfy $c_i/s < d_i < c_i$. The d_i increase strictly with c_i , whereas the δ_i decrease strictly with c_i .

Proof. Equation (4) gives $DA^{-1}p(c) = Dp'(c) = p(c)$; the vector is nonzero under the assumptions. The formulas in (13) follow by logarithmic differentiation. The bounds and strict monotonicity follow directly from the positive summands in those formulas. $\square$

The lemma supplies one eigenvalue, not an entire repeated spectrum. There is no claim that every solution admits the negative-root representation (12).

3 Complete existence for two stages

Theorem 3.1. *For any $0 < a < b$, set $r = \sqrt{2ab}$. For nodes $c_1 = a, c_2 = b$, the diagonal*

$$d_i = \frac{c_i(c_i + r)}{2c_i + r}, \quad i = 1, 2, \quad (14)$$

is positive and satisfies $(I - D^{-1}A)^2 = 0$.

Proof. Apply Lemma 2.2 to $p(t) = t(t + r)$. One eigenvalue is one. Moreover,

$$2\delta_1\delta_2 = \frac{2(a+r)(b+r)}{(2a+r)(2b+r)} = 1,$$

since $r^2 = 2ab$. Thus $\det(A^{-1}D) = 1$ and the other eigenvalue is also one. Proposition 2.1 completes the proof. $\square$

There is no root search in this construction. The excluded one-stage case is also immediate: $A = [c_1]$, $D = [c_1]$.

4 Complete existence for three stages

This is a constructive proof for *every* positive distinct triple, not a statement restricted to particular quadrature nodes.

Theorem 4.1. *Let $0 < c_1 < c_2 < c_3$. There exist $r > 0$ and $q > 1$ such that*

$$d_i = \left(\frac{1}{c_i} + \frac{1}{c_i + r} + \frac{1}{c_i + qr} \right)^{-1} \quad (15)$$

satisfies $(I - D^{-1}A)^3 = 0$. In this construction, $d_1 < d_2 < d_3$ and $c_i/3 < d_i < c_i$.

Proof. Write $h_i = B_{ii}$ and $e_1 = c_1 + c_2 + c_3$, $e_2 = c_1c_2 + c_1c_3 + c_2c_3$, $e_3 = c_1c_2c_3$. For $q \geq 1, r \geq 0$, define

$$\delta_i(r, q) = \left(1 + \frac{c_i}{c_i + r} + \frac{c_i}{c_i + qr} \right)^{-1}.$$

For each fixed finite q , each δ_i increases strictly from $1/3$ to 1 as r increases from zero to infinity. Consequently there is a unique $r(q) > 0$ satisfying

$$6 \prod_{i=1}^3 \delta_i(r(q), q) = 1. \quad (16)$$

The derivative with respect to r is strictly positive, so $r(q)$ is continuous (and smooth). Put

$$T(q) = \sum_{i=1}^3 h_i \delta_i(r(q), q).$$

We show that $T(1) < 3$, whereas $\lim_{q \rightarrow \infty} T(q) > 3$.

A rational trace identity. We have $\sum_i h_i = 6$. Pairing the terms with indices i, j gives, for $x > 0$,

$$\begin{aligned} \sum_i \frac{h_i}{c_i + x} &= \sum_i \frac{1}{c_i + x} + \sum_{i < j} \frac{x}{(c_i + x)(c_j + x)} \\ &= \frac{6x^2 + 3e_1x + e_2}{x^3 + e_1x^2 + e_2x + e_3} > 0. \end{aligned} \quad (17)$$

This identity also avoids any assumption that individual h_i are positive.

The endpoint $q = 1$. Here $r = r(1)$ and $\delta_i = (c_i + r)/(3c_i + r)$. Let

$$P(r) = \prod_i (c_i + r), \quad R(r) = \prod_i (3c_i + r).$$

Using (17) with $x = r/3$, or expanding directly, yields

$$R(r)(T(1) - 3) = 6P(r) - R(r) - 2(r^3 + 3e_3). \quad (18)$$

At $r = r(1)$, (16) says $6P(r) = R(r)$. Therefore

$$T(1) - 3 = -\frac{2(r(1)^3 + 3e_3)}{R(r(1))} < 0.$$

The endpoint $q \rightarrow \infty$. For fixed $r > 0$, $\delta_i(r, q)$ increases with q and tends to $(c_i + r)/(2c_i + r)$. Thus $r(q)$ decreases with q . The function

$$\prod_i \frac{c_i + r}{2c_i + r}$$

is strictly increasing from $1/8$ to 1 , so there is a unique $r_\infty > 0$ for which this product equals $1/6$. For every finite q , $\delta_i(r, q) \leq (c_i + r)/(2c_i + r)$; hence $r(q) \geq r_\infty$. It follows that $r(q)$ has a positive limit. Passing to the limit in (16), using $qr(q) \geq qr_\infty \rightarrow \infty$, identifies that limit as r_∞ . Equation (17) with $x = r_\infty/2$ now gives

$$T(\infty) - 3 = \frac{3r_\infty^3 + 3e_1r_\infty^2 + 2e_2r_\infty}{\prod_i(2c_i + r_\infty)} > 0. \quad (19)$$

Closing the spectral equations. Continuity supplies a finite $q_* > 1$ with $T(q_*) = 3$. Set $r = r(q_*)$ and choose D from (15). The polynomial $p(t) = t(t + r)(t + q_*r)$ gives one eigenvalue equal to one by Lemma 2.2. Equation (16) gives determinant one, and the trace is three. If

$$\det(\lambda I - A^{-1}D) = \lambda^3 - 3\lambda^2 + \sigma_2\lambda - 1,$$

substitution of $\lambda = 1$ gives $\sigma_2 = 3$. Thus the characteristic polynomial is $(\lambda - 1)^3$. Proposition 2.1 gives the target. The bounds and ordering follow from Lemma 2.2. $\square$

4.1 A terminating scalar-bracketing construction

For a chosen q , bracket $r(q)$ by doubling an upper bound until $6 \prod_i \delta_i(r, q) > 1$, and then bisect. Start the outer search at $q = 1$ and double $q = 2, 4, 8, \dots$ until $T(q) > 3$; then bisect that outer bracket. The strict endpoint inequalities prove that finite brackets exist in exact arithmetic. Uniqueness of the outer root is neither needed nor asserted.

For example, for $c = (1/3, 2/3, 1)$, the reconstructed computation gives

$$q \approx 14.68040223705983745, \quad r \approx 0.27135557452443663,$$

$$d \approx (0.20469201760073983, 0.35956816768189391, 0.50321559543241252).$$

These displayed decimals are approximations to the construction, not a replacement for the preceding proof. The provided implementation checks brackets and precision and raises errors rather than declaring failed searches to be counterexamples.

5 An all-stage local existence theorem

The following result is uniform over nearby node sets for any fixed stage count. It is not restricted to stages two and three, but it is local in node space.

Theorem 5.1. *For every integer $s \geq 2$, there exists $\varepsilon_s > 0$ such that the following holds. For any $\tau > 0$ and distinct positive nodes satisfying*

$$\left| \frac{c_i}{\tau} - 1 \right| < \varepsilon_s, \quad i = 1, \dots, s,$$

there is a positive diagonal D satisfying (2). A branch of such diagonals can be chosen real analytically in the nodes near the coincident-node configuration. Along this branch,

$$d_i \longrightarrow \frac{\tau}{(s!)^{1/s}} \quad \text{as all } c_i \longrightarrow \tau. \quad (20)$$

5.1 A polynomial extension across coincident nodes

By scale invariance it suffices to work near $\tau = 1$. Let

$$\beta_b(z) = \sum_{j=1}^s b_j z^j, \quad g_b(t) = -\beta'_b(1-t).$$

When $g_b(c_i) > 0$, take $d_i = 1/g_b(c_i)$. For distinct nodes define

$$F(c, b; \lambda) = \frac{\det [kc_i^{k-1} - \lambda g_b(c_i) c_i^k]_{i,k=1}^s}{s! \prod_{i < j} (c_j - c_i)}. \quad (21)$$

Set $E = (c_i^k)_{i,k=1}^s$ and $E' = (kc_i^{k-1})_{i,k=1}^s$. Since $E(E')^{-1} = A$, factoring the numerator gives

$$F(c, b; \lambda) = \det(I - \lambda D^{-1} A). \quad (22)$$

Each numerator entry in (21) is polynomial in the nodes, b , and λ . The numerator is alternating in the nodes, so it is divisible by their Vandermonde product. Consequently F extends as a polynomial in all its arguments to coincident nodes. Its constant term in λ is always one. This auxiliary extension is well defined even though the ordinary Lagrange collocation matrix is not defined at coincident nodes.

Lemma 5.2. *At $c = (1, \dots, 1)$, the polynomial extension satisfies*

$$F((1, \dots, 1), b; \lambda) = \sum_{j=0}^s [z^j] \exp(\lambda \beta_b(z)). \quad (23)$$

Here $[z^j]$ denotes coefficient extraction; only finitely many Taylor coefficients of the exponential are used.

Proof. The confluent limit of a determinant divided by a Vandermonde determinant is the determinant of Taylor coefficients of its column functions at the common point. This follows by expanding every row in a Taylor series: the lowest nonzero alternating term uses derivative orders $0, 1, \dots, s-1$, and its coefficient is the Taylor-coefficient determinant.

Equivalently, the numerator in that limit represents, on $\mathcal{P}_s^0$, the first s Taylor coefficients at $t = 1$ of $p'(t) - \lambda g_b(t)p(t)$. Write $q(z) = p(1-z)$; the condition $p(0) = 0$ is $q(1) = 0$. The homogeneous Taylor equations become

$$q'(z) - \lambda \beta'_b(z) q(z) = 0 \pmod{z^s}.$$

Writing $q(z) = \sum_{j=0}^s a_j z^j$, these equations determine $a_1, \dots, a_s$ successively from a_0 , with constant diagonal coefficients $1, 2, \dots, s$. Their solution is

$$q(z) = a_0 \exp(\lambda \beta_b(z)) \pmod{z^{s+1}}.$$

The remaining equation $q(1) = 0$ has coefficient $\sum_{j=0}^s [z^j] \exp(\lambda \beta_b(z))$ on a_0 . Gaussian elimination in this augmented system therefore produces this factor times the constant factor $s!$, up to basis signs independent of λ . Dividing by the determinant at $\lambda = 0$, exactly as in (21), removes those constants and proves (23). $\square$

5.2 A triangular solvability map

Write the right side of (23) as

$$1 + \sum_{k=1}^s f_k(b) \lambda^k, \quad f_k(b) = \frac{1}{k!} \sum_{m=0}^s [z^m] \beta_b(z)^k.$$

We seek $f_k(b) = (-1)^k \binom{s}{k}$. Put $a = (s!)^{1/s} > 0$ and choose

$$b_1 = -a. \quad (24)$$

Since $f_s = b_1^s/s!$, this solves the highest-degree coefficient equation. For $j = 2, \dots, s$, let $k = s - j + 1$. Degree counting in β_b^k gives

$$f_k(b) = \frac{b_1^{k-1}}{(k-1)!} b_j + R_j(b_1, \dots, b_{j-1}). \quad (25)$$

Indeed, a term containing $b_j z^j$ can survive through degree s only by using $k-1$ copies of $b_1 z$, giving total degree $j + k - 1 = s$. Two copies of b_j have degree greater than s , and any b_m with $m > j$ also gives degree greater than s .

Thus the remaining coefficients are obtained uniquely by the finite recurrence

$$b_j = \frac{(k-1)!}{b_1^{k-1}} \left((-1)^k \binom{s}{k} - R_j(b_1, \dots, b_{j-1}) \right). \quad (26)$$

Call the resulting real vector b^* . The Jacobian of the map $b \mapsto (f_s, f_{s-1}, \dots, f_1)$ is lower triangular, and

$$\det \frac{\partial(f_s, f_{s-1}, \dots, f_1)}{\partial(b_1, b_2, \dots, b_s)} = \frac{b_1^{s(s-1)/2}}{\prod_{j=0}^{s-1} j!} \neq 0 \quad \text{at } b = b^*. \quad (27)$$

The first diagonal entry is $b_1^{s-1}/(s-1)!$; the others follow from (25).

Completion of the proof of Theorem 5.1. For general c , form the s nonconstant coefficients of $F(c, b; \lambda) - (1 - \lambda)^s$. This is a real polynomial map. At $(c, b) = ((1, \dots, 1), b^*)$ it vanishes, and its derivative in b is invertible by (27). The real analytic implicit-function theorem therefore supplies a real analytic solution $b = b(c)$ for all c in a neighborhood of $(1, \dots, 1)$.

At the center, $g_{b^*}(1) = -b_1^* = a > 0$. After shrinking the neighborhood, continuity ensures every $g_{b(c)}(c_i) > 0$. Define $d_i = 1/g_{b(c)}(c_i)$. For distinct nodes, (22) gives

$$\det(I - \lambda D^{-1}A) = (1 - \lambda)^s.$$

Hence the characteristic polynomial of $D^{-1}A$ is $(\mu - 1)^s$, and Cayley–Hamilton proves (2). The diagonal tends to $1/a$ at the center. Applying scale invariance establishes the assertion for arbitrary τ , including (20). $\square$

The recurrence and Jacobian prove a nonempty open region of solvability in every dimension. This proof does *not* provide an explicit value of ε_s , or justify continuation of this branch to arbitrary separated nodes. No positivity of the coefficients b_j beyond the chosen sign of b_1 is needed.

6 An exact four-stage certificate

For $c = (1/4, 1/2, 3/4, 1)$, scale invariance lets us calculate B from the nodes $(1, 2, 3, 4)$:

$$B = \begin{pmatrix} -5/6 & -1 & -1/2 & -1/3 \\ 2 & 0 & -2 & -1 \\ 3/2 & 3 & 5/2 & -3 \\ 4/3 & 2 & 4 & 25/3 \end{pmatrix}.$$

The exact certificate is stored in

`results/four_stage_certificate.json`

It defines rational polynomials $u_1(z), \dots, u_4(z)$ and a rational isolating interval for a real root α of the following polynomial:

$$\begin{aligned} p(z) = & 39343011840000z^{14} - 240866661376000z^{13} \\ & + 648960650342400z^{12} - 1039122239554560z^{11} \\ & + 1122434419969024z^{10} - 860708601711360z^9 \\ & + 481628037063168z^8 - 198176172858240z^7 \\ & + 59544616187520z^6 - 12888571783800z^5 \\ & + 1965366900400z^4 - 203649946500z^3 \\ & + 13930702500z^2 - 674415625z + 23765625. \end{aligned}$$

Here $u_3(z) = z$, and $\alpha \approx 0.42565199554545926759221216771143468$. The remaining parameter polynomials have large rational coefficients and are stored in descending coefficient order in the certificate, rather than printed as several pages of integers.

The verification program performs three exact operations. It counts exactly one real root of p in the stored rational interval. Rational interval Horner evaluation proves $u_i(\alpha) > 0$ for all four coordinates. Finally it checks that, for each $k = 1, 2, 3, 4$, the polynomial

$$\sum_{|J|=k} \det B[J, J] \prod_{j \in J} u_j(z) - \binom{4}{k}$$

has zero remainder on division by $p(z)$. These facts prove that $\delta_i = u_i(\alpha)$ is a real positive solution of the full system (9). No floating-point residual threshold enters this certificate. The diagonal is $d_i = (i/4)u_i(\alpha)$, approximately

$$d = (0.131746889029651, 0.229321788436070, 0.319238996659094, 0.405003156948222).$$

Generation uses a rational Gröbner-basis computation, but replay does not need to trust the search that found the certificate: it independently checks the isolated root, positivity, and all four equations. The conclusion is an exact existence statement for this specified node set, not an arbitrary four-stage theorem.

7 Recomputed experiments and validation limits

The archive contains `results/numerical.json`, generated with 110 decimal working digits. Each candidate is checked using

$$R_{\text{coeff}} = \max_{1 \leq k \leq s} \left| \binom{s}{k}^{-1} \sum_{|J|=k} \det B[J, J] \prod_{j \in J} \delta_j - 1 \right|$$

and the direct quantity $\|(I - D^{-1}A)^s\|_\infty$. The latter matrix norm is the maximum absolute row sum. Results are rounded for display below.

Nodes	R_{coeff}	$\ (I - D^{-1}A)^s\ _\infty$
(1/100, 1)	$8.32 \cdot 10^{-111}$	$5.68 \cdot 10^{-111}$
(1/3, 2/3, 1)	$5.46 \cdot 10^{-50}$	$1.40 \cdot 10^{-48}$
(1/100, 1/10, 1)	$9.52 \cdot 10^{-51}$	$1.44 \cdot 10^{-49}$
(4/5, 9/10, 1)	$5.86 \cdot 10^{-49}$	$3.45 \cdot 10^{-46}$
$i/4, \quad i = 1, \dots, 4$	$4.99 \cdot 10^{-110}$	$6.80 \cdot 10^{-108}$
$i/5, \quad i = 1, \dots, 5$	$1.16 \cdot 10^{-110}$	$1.89 \cdot 10^{-107}$
$i/6, \quad i = 1, \dots, 6$	$7.32 \cdot 10^{-110}$	$1.73 \cdot 10^{-105}$
$i/7, \quad i = 1, \dots, 7$	$1.50 \cdot 10^{-110}$	$2.54 \cdot 10^{-105}$

The two- and three-stage rows approximate constructions justified above. The four-stage node set additionally has the independent exact certificate. The five-, six-, and seven-stage rows are numerical evidence only. The working precision is not a claim that every displayed or stored decimal is certified to that many places. In particular the nested three-stage bisections used a substantially looser requested tolerance.

The symbolic regression script checks the two-stage identities, both general three-stage endpoint identities, exact instances of the Cauchy similarity, and the confluent determinant and Jacobian formulas for stages one through four. These finite symbolic tests supplement the arbitrary-stage proof in Section 5; they do not establish it by extrapolation.

Searching the characteristic coefficients rather than eigenvalues avoids using a particularly sensitive numerical quantity as the primary test. For example, the companion matrix of $\lambda^s - \eta$ tends to a nilpotent Jordan block as $\eta \rightarrow 0$, yet its eigenvalues have modulus $|\eta|^{1/s}$. Thus an extremely small coefficient perturbation can cause appreciably larger apparent eigenvalue splitting. Conversely, small computed eigenvalues by themselves do not certify an exact repeated-root identity.

8 What is and is not established

The proven assertions are: full existence for arbitrary two-node and three-node sets; local existence near a common positive node in every dimension; exact equivalent equations in every dimension; and an exact positive solution for the specified equally spaced four-node example.

The missing assertion is existence for *every* admissible node set in *every* stage count, beyond the regions and cases just proved. The local implicit-function argument does not rule out a singular continuation Jacobian, loss of positivity, or escape from a bounded parameter set away from its initial neighborhood. The total-positivity and relative-gain-array ideas explored in the interrupted notes supplied no verified lemma that removes those possibilities. No global claim based on them is included. Likewise, a failed nonlinear solve is not a certified counterexample.

Accordingly, this is a recovered, checked partial research package, not a resolution of the universal IE-28 conjecture.

Reproduction

From the archive root, a Python environment with the two packages in `requirements.txt` is sufficient:

```
python -m pip install -r requirements.txt
python code/check_symbolics.py
python code/exact_certificate.py
python code/run_experiments.py
```

To regenerate the algebraic search as well as replay its certificate:

```
python code/exact_certificate.py --generate
```

To rebuild this document, run `pdflatex report.tex` twice or `latexmk -pdf report.tex`. The numerical and exact-check logs included in the archive record runs completed during reconstruction.

References

- [1] OpenProblemsInNLA repository contributors, *Open Problems in Numerical Linear Algebra*, entry IE-28, “Positive diagonal preconditioning with a nilpotent stiff limit.” Problem page accessed 12 September 2026. <https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/main/linear-systems-and-elimination/IE-28/README.md>.
- [2] P. J. van der Houwen and J. J. B. de Swart, “Triangularly implicit iteration methods for ODE-IVP solvers,” *SIAM Journal on Scientific Computing* **18**(1) (1997), 41–55. See Section 3.2.1, p. 46, and Section 3.2.2. <https://ir.cwi.nl/pub/2191/2191D.pdf>.
- [3] G. Čaklović, T. Lunet, S. Götschel, and D. Ruprecht, “Improving efficiency of parallel across the method spectral deferred corrections,” *SIAM Journal on Scientific Computing* **47**(1) (2025), A430–A453. See Section 2.2.3, p. A439. <https://d-nb.info/1363153935/34>.